Discussion with Michael Pollan of new ideas on memories and Selves

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great so this is Mike L on um Wednesday May 8th from home it looks like I'm I'm at home today yeah yeah me too me too well I thought this sales piece was absolutely fascinating just oh thank you fireworks of ideas coming off it just very rich um so I'd love to ask you a few questions about it just to unpack it and then we can take it from there um first of all connect this thinking and this piece to your lab work what what what brought you to this these thoughts and what yeah what were you seeing that made you want to answer the questions you answer in this piece yeah great question uh let's see a couple of things first um I am really interested in information and how information is is used by different components of the living body so you know Josh Josh bongard and I have been developing this notion of polyc computing which is this idea that basically every subcomponent is hacking every other subcomponent in the sense that it doesn't know or care what the original intent of the message was it's going to interpret it as best as it can how you know however it can when you say it are you talking about organisms you what are you talking about well all every every level so cells subcellular protein networks Pathways organs tissues you know anils every every everything right I think I think this is a fundamental feature of biology that it's a hacker in that sense that that you have no allegiance to how something was intended to be used you're going to do the best you can in interpreting what and and and and that comes from from some you know working with Josh bongard with with his um discoveries that the exact same physical mechanism can be seen by one Observer to be Computing one function and another Observer to be be Computing a different function so that that right that when when you ask you know what is this physical process Computing somebody might say well I know what it is I wrote the algorithm so I can tell you what it is but in by biology nobody cares who wrote the algorithm well yeah well I don't care what you say I'm interpreting your your algorithm your novel your message in my own way whatever I I you know it doesn't matter to me what what you think you meant by it I get whatever I get out of it is what I get out of it right so so uh anyway so so it's so it starts with that and and attempts to understand how different uh biological systems uh interpret information and I was thinking uh okay so that's one thing the second thing thing is I I really think that there is a lot of important work to be done to Break um binary categories that we naively think exist but actually are just hiding a lot of um limits on our imagination and so this this notion of data to data versus the machine that operates on that data and this idea that information is passive and then you have this active uh a cognitive being whatever it is that's going to operate on that data remember it you know store it change it you you know what whatever and so so I so I was thinking about um ways that that we could we could break that and really make that into a Continuum and I was thinking about the whole but caterpillar butterfly thing and I thought about this for for for you know many many years uh and the originally the thing you might think is cool about it is the question of where's the information because the brain gets mostly refactored okay but then it hit me I was I was literally I was I was taking a walk with with my dad and it and it and I was thinking about this and it hit me that that isn't the most interesting part here what's interesting is that the actual the the the details of the information that the butter that the caterpillar learns are actually quite useless to the butterfly because it's going to have to remap all of it onto a completely different body with different priorities it doesn't care about leaves anymore it you know it lives in threedimensional world like all of that has to get remapped and so immediately I sort of I sort of realized that what's going on here is is is the what's preserved across lifetimes from from that caterpillar to that butterfly is not the f it of the information it's the it's a kind of um inferred salience it's it's what does this mean to me and that what's actually passed on physically some sort of engram we don't know what it is it could be an RNA like a glanc says what's an engram would you define that term for me yeah an engram is just a it's a it's a physical embodiment of a memory so so it's it's anything in your brain or body that uh stores information and all that means is that there is some Observer later on meaning a cell a tissue the whole animal or or a scientist or somebody else who's going to then look at that physical object and say oh look I know what this means this is going to interpret this as as as as a memory and so you know ant ants is DNA in engram or not yeah I think so yeah okay yeah yeah I think so but now now again that that requires that requires a real shift I think most biologists would say no but I think it is because I think that um all memories are just messages from your past self and that what's happening with DNA is that previous previous basically there's this giant lineage agent that's the scale of an evolutionary lineage and the DNA are its engrams where that information is is being passed on the way that any memory would and much like these other memories it's up for interpretation which is why you have this this incredible flexibility of forms yeah you've got the same DNA but if you're a salamander and I changed the number of cells or the size of cells or or you know do all this other crazy stuff you can still make things work because you don't interpret you take it seriously but not literally that information so right so yeah so so going back to the butterfly yeah yeah yeah so so the engrams right um that uh that that are left there are to be interpreted and that leads to this this question of uh and again the reason so get get back to your original question what does it mean for our lab work because because what we would like to do is communicate with effectively communicate with the different parts of the biology I mean what we do is we communicate with cells tissues and organs for regenerative medicine applications some people are going to want to communicate with ecosystem somebody else is going to want to communicate with you know uh uh so social structures or or financial structures I don't know what and so so we're interested in asking how does you know by by understanding how does a persistent agent a self maintain itself um over time you can learn something about how do we communicate with it so we would like to rewrite those memories and we would like to say if there's a birth defect or there's a uh some kind of traumatic injury or something and we want the cells to build something else we really need to understand how these memories are interpreted so that's why yeah that's that's that's why this this is of relevance to us so in the butterfly case would you say there is a self that's continuous from from caterpillar to butterfly so so here's what I think we ought to do with this and this is by no means a new a new idea this is this is kind of just process philosophy in many made made more practical in many ways um you know there's a there's an old Paradox and and I'm trying to figure out who first said it I I thought it was biton but I could be wrong but but somebody had this Paradox that said if if a species fails to change then it will no doubt eventually go extinct if it does change then again it's not the same species and so again it's gone so what do you what do you do what's the right what can you do that's the Paradox and and that same Paradox is um it's it's it's it's the same thing with with us because in a certain sense if you if you insist on a definition of the self that is a permanent structure it's a thing then you've got the same Paradox because if I mature if I learn if I expand if I you know any of those things I'm no longer me and that's scary and at some point you know is the child that you once were still here I mean in many ways no right so that's a that's a problem but I think I think the answer both on the personal case and the uh evolutionary case is that what we need to do is Define the self as a continuous construction of sense making so it's a process it's not a single thing it's a so so that part isn't new plenty plenty of people have said that but but I I think the new thing is to understand exactly what it's doing what I think it's doing is constantly trying to make sense of its own memories so every and I don't know how what 300 milliseconds you know in a human I don't know what it what it would be exactly but you you don't have access to your past what you have access to is the engrams left by your past you have the messages from your past celf so this constant process of trying to make sense of what do my what do these chemical um Messengers mean if it's RNA or it's prot you know whatever uh synaptic structures whatever it's going to be uh what do they mean can I can I Cobble together a coherent story of myself my past my environment and be creative about it the the the the thing the the other thing that hit me that's I think really cool about this is that um uh and we could talk later about this this bow tie business but but in any case what's happening is you have you have a bunch of experiences and you don't memorize the details you have to squeeze them down you have to compress them into some sort of a inference a generative you know a model that's of of everything that happened right so uh when you when it comes time to to interpret this thing to recall the memory because because it's compressed you have to sort of reinflated it and you have to make it make sense for your new environment maybe you you're physically you've changed or whatever you have to you have to reinl it but the thing is that good compression always looks like randomness you know this is something that the seti people uh point out that that a really Advanced signal is going to look maximally random because because when you compress lots of particulars into a general rule you the whole point of compression is is to throw out all the correlations anything that's correlated you can get rid of it because you can comp you know you can compress it out but once you've done all of that what you have looks really random and so that means that on the interpretation end when you look at these molecules or synaptic structures or whatever it is in order to uh figure out what they mean you can't deduce it in a deductive algorithm because the information isn't there it's been it's been stripped off you have to be creative it's it's more it's more creativity and uh you know um um maybe what people call intuition I don't know but it's but it's much more of a you you have to bring something to it you can't just read exactly what it says because right it's sparse there's not there's very little there why now tell me exactly why do you have to compress you have to compress bandwidth question well because because if you don't compress all learning is about compression because because if you don't compress then what you've done is what what the machine science machine learning folks call um overtraining you've you've you've memorized a bunch of particulars and if and and if you're a very simple organism in a very simplified environment that might even be okay but but normally the whole point of learning is you learn a rule you don't remember there were these pixels on my retina then there were those pixels on my R you don't remember that what you remember is this General thing leads to this other General thing right you remember these these connections abstracting exactly that's what it is it's generalization and abstraction that's exactly and that seems to me what's happened in the caterpillar butterfly case right I mean so in the experiment that you describe it wasn't I realized it wasn't yours though it was somebody else's experiment yeah yeah yeah it's not it's not my experiment it was done it was first done in the 70s um by a bunch of Russian groups who did uh um larv Beetle and things like that and then it was done by uh Doug Blackiston who is actually a staff scientist in my center but but he did this work when he was younger and uh yeah he did it with moth caterpillar to moths and he did it there was some training some Operating Conditioning going on right to associate food with a certain color yep and so the abstraction there or the result of the compression was concept food right rather than specific leaf or nectar well it's two th right it's two it's actually multiple things because a it's yes we're not remembering leaves because butterfly doesn't care about leaves but also uh what do you do because because um the the caterpillar is a soft body creature and the way you move a soft body you don't have any hard elements to push on so you can't do robotics the same way you can't do controlling the same way so there's a certain set of muscular motions that you need to do to make your way over to where the where the leaves are that isn't going to help the butterfly the butterfly doesn't move like that it's got a completely different architecture and and so so they have the abstraction there is from from uh from leaves to food or maybe even we don't know it might even just be to um pleasure or something you know just just positive affectors you know who knows right um but also but also how do we make use of that associative learning that this color leads to food or to you know to to to pleasure or whatever um how do we remap that on a completely different controller and and you see this even in vertebra it's like like this is um and actually this was Doug's Doug's work to is uh we can make tadpoles with eyes on their tails and they can see perfectly well and so it's like isn't that amazing that with no extra evolutionary Cycles no no adaptation out of the box this this thing that for millions of years had the exact you know the brain could rely on getting connection from the visual system into the optic tectum and all that H Suddenly It's on your tail no problem it you know connects to the spinal cord didn't even connect to the brain it makes it work so what's happened there do you think how do you think that what's been remapped what's been remembered and forgotten I I think that all of these and there's tons of these examples that I go through in various papers is the reason that um life is so tolerant to these crazy kind of changes is that it fundamentally uh assumes the material is unreliable in other words unlike all of our computer information the information the substrate it it assumes that the substrate is unreliable because look at this this is how we build our computer technology you make the bottom layer as absolutely rock solid as you possibly can so that all of the stuff on top of it can assume that everything is going to go well at the bottom you know when you write code you don't think your your your um transistors is going to burn out you don't worry about that and so so the goal is to change the to to to to keep the information from changing and this is why actually that whole business of these microchips right how small can you make a microchip it it's because you don't want the bits floating when they get too close to each other they start to fluct you know the quantum effects start to fluctuate you don't want that you want every you want all the bits to be exactly what they were before you want Fidelity of the information biology can't you can't operate that way because everything is going to change so on a small time scale your proteins are going to degrade your cells are going to die and be replaced on a larger scale you're going to be mutated right Evolution you know guaranteed you're going to be mutated and things will things will change not just environment but also your own parts so um I think that there there may be Exceptions there may be organisms that aren't like that although I doubt it I think at this point what what survived and what what um biology uh strongly um emphasizes is architectures where you assume the underlying material is unreliable and what you're going to do is on the Fly you don't overtrain on your priors you don't assume that the future is going to be like the past you don't assume you know what any of your information means you are going to try to reinterpret it at the at any given moment on the Fly and do the best you can and so that gives rise to uh these kinds of systems where the eyes in the wrong place we can we can do something with that basically right I mean the detail is is it's it's hard work to figure out okay so um where is the information actually going you know it gets onto the spinal cord then the brain learns to infer some connection between what the eye says and what the does it even know it's an eye I'm not even sure about that so but I but I think that's what's happening here is that um biology commits to a realtime sens making process not to the expectations of the past you know I think I think that's the most interesting part and that's what makes the stuff so flexible and that's what makes bio biotechnology possible it makes you know I'll take my cells and slap them onto some weird scaffold made out of crazy nanomaterials and I'll instrumenti it with electrical signaling and Optical whatever and things work because you know so the so the the butterfly has solved this problem of um how can I change and uh and still can I change and still exist which I I think I think we all do I think I mean the butterfly is kind of an egregious case but but um all of us have the same problem because over time we are not the same and you know whether it be the rearrangements of of of puberty or just the the you know the day-to-day wear and tear on the body you know the the the whole ship of thesias thing right where where your your materials are going in and out all the time um I think I you know I think I think different species emphasize this to different extent for example plaria are the champions of this I mean look planaria right uh cancer resistant incredibly regen regenerative uh no Aging in the asexual forms right no transgenic lines because they basically ignore new DNA that you put into them there are no transic yeah there's no for people have been trying this since uh I think since the late 80s people have been trying to make transgenic planaria there aren't any no no mutant lines the only the only lines of aberant plaria that exist are our two-headed form and the cryptic form and they're not genetic it's because the only reason it works is because it's not genetic uh what happened in uh in plenaria I think is that they uh because because they're they reproduce asexually so they you know tear themselves in half and regenerate that means that they accumulate sematic mutations they don't clean the genome the way that that we do with sexual reproduction and so they've accumulated so much junk that basically the only way to to have a proper planarian is to assume the hardware is is going to be unreliable and to put all of the effort and we've done computational models of this actually and you can see Evolution doing this when when you get even a little bit of um regulative competency where the cre where the creature can make up for certain subtle defects immediately Evolution you it's hard for selection to now see the genomes right because if you get an animal that looks pretty good do you look pretty good because your genome was great or because you actually it's terrible but you're f fixed it along the way right and so this is like the stuff that we see when we make Tates with the with the scrambled faces and they see them they they they fix them right so so what happens is that uh then when Evolution has a hard time uh selecting for good genomes all the effort goes into selecting for more competency which in turn makes it harder to see the genome which in turn makes for more and so you get this you get this thing where the the pressure on the genome actually flattens out but the pressure on the competency keeps rising and so if you take that all the way way you end up with plaria that basically everything went into making an algorithm which is partially bioelectric and who knows what else that basically says I already know my Hardware can't really be trusted here are all the error correcting codes and everything else that we need to do to build a good planera no matter what happens and then of course you're insensitive to trans genes to you know to to cancer all these things because you're you're you're you're assuming from day one that all that stuff can't be trusted right right you know and I think you know I think I think so plenaria are kind of all the way there and then salamanders are pretty good at it but not as good as plenaria and then then mammals and then maybe something like sea elegans or drop are on the opposite end and they're just really pretty you know pretty hardwired maybe uhuh so your when you your definition of self how far down the evolutionary ladder does It Go I mean does it go to single-cell creatures or they selves or does it uh go or does it begin at a certain point in evolution the self as an innovation to to deal with these problems you're talking about yeah um I don't I I don't like um binary categories for any of these things because they end up chasing us into these pseudo problems where you can always come up with these sort of in between cases and then you spend all your time trying to prop up this binary definition as opposed to I I I think I think what's what allary here between having a and not having aelf correct yeah yeah exactly it's on a Continuum I I think all of it well well yes I think it's on a Continuum but but the reason it's on a Continuum is that I take all of these terms so all of this you know having a self intelligent sensient you know cognitive like all the whatever of these terms you want to use I don't think they're about the system itself I think what they refer to is your intended interaction with it so if you tell me that you think a certain system as an observer as a scientist as a okay as as as a scientist as a as a conspecific as a parasite as a you know whatever whatever it's it's every every agent with living non-living scientific you know natural whatever you are going to take some stance towards whatever you want to interact with and if you want to take a mechanical stance and say all I see is a bunch of cogs and gears I'm you know my the tools I have to interact with you are just Hardware rewiring well you know that works well for mechanical clocks and things like that you try to apply it to a human if you're an orthopedic surgeon not bad if you're a psychotherapist terrible right so so you know you you want you want an or or or or a spouse you know terrible you you want an orthopedic surgeon that thinks you're a mechanical machine you do not want dispose or or a psychotherapist that thinks you're a mechanical machine so I think all of these things um I think all of these things uh just just indicate the frame that you bring to the interaction and the set of tools you're going to use and so you got you know you got your rewiring you've got cybernetics control theory you know behaviorism you've got I don't know psychoanalysis spirituality who know like I'm just you know sort of drawing the Spectrum right so so how far down down does it go so uh that so so what that means is can we get some sort of utility and that's why I I harp on the engineering side of this because I think this should all be tested by by utility in the in the real world so so can we get some sort of utility by applying these Concepts to single cells absolutely I think yes um can we apply them to molecular networks yes within cells I think so we have we have data on this other people have data on this could we apply them to uh particles maybe so I think um Chris fields and Carl friston and some other people have done some really nice work trying to cash out physics as a kind of uh protoc ogni um substrate like active I mean they like active inference there may be some other things um so maybe so so I was going to ask you too if you thought this was limited to the realm of biology I realized friston doesn't um yeah but I don't know he does his his work makes a better seems to all perform better when you transpose it to the realm of biology then when he starts talking about active inference and rocks and crystals and things um but I don't see the active piece um well I so so okay yeah you know I I couldn't possibly repr reproduce the argument the way the way that he does it but but I think I think what he's saying is that um and and and Chris Fields has has stories about this too which is basically that there's an equivalency between um what what is what is a thing you know what is a rock what is a right what I mean I think that's a very important part of the research program and this idea that there's the active inference is quite symmetrical when you're doing something to the environment the environment is also learning about you and and so on and I think the claim there is that there is a very simple version of this that just basically looks like physics to us but if you sort of crank up the the the relevant parameters then you end up with things we call life so so so here's here's what I would say about that um I I I think when we say biological world what do we really mean right what what what's life I I think that what we mean by life is anything that is good at scaling up the cognitive light cone that is anything that um where the hole has a larger uh is capable of pursuing larger and more complex goals than the parts that's what we happen to call life so so we don't do that for rocks because The Rock has exactly the same cognitive light going as as the pieces that go inside rock it hasn't scaled anything it's just you exactly the same it's not more than the sum of its parts in in in that particular way right so so some people would do it you know Toni would do it off of um integrated information different people do it different ways I I think it's all about gold directedness and right I I think it doesn't it doesn't uh you know you're not going to find um larger goals that help you deal with the rock it's basically the same it follows least action principle that's about all you're going to get out of it right right so right so so I think I you know think biology is what we roughly call things that are good at it and that again is a

Continuum I think we don't need to spend any time arguing whether something is alive or not the real question is so what's so so what's your model what what goals do you think this follows and how does that help you have a richer interaction with the system and and what tools can you bring from from active influence from behavior science and and and so on I mean look one one example okay one one recent example is we showed models of Gene regulatory Network which are just chemicals you can might as well be a rock it's just you know six or seven chemicals interacting with each other and what we showed is that if you bring tools from from Behavior theory meaning meaning different kinds of learning so associative conditioning habituation sensitization you can do some really interesting things with just with those networks that have lots of biomedical uh uh relevance and and and you know it's already there you don't you don't need a cell you don't need any you know you don't need protoplasma you don't need any of that just just the mathematics of having a few uh nodes connected by these differential equations already give you a bunch of stuff that we would call learning and that help you you know and and and again my goal right remember you know my goal is not to to do poetry and to sort of paint hopes and dreams onto these things and and uh you know that we could never agree on I I make a very clear claim that if you know these things you will do better in the biomedical Arena than if you pretend they don't exist and that's that's a you know that's not a philosophical claim that's that's an empirical claim that we can you know that we test you mean as a scientist or as a creature no as uh well both uh so so first and is a so so mostly what I talk about in public is the science right so so I say as as a worker in regenerative medicine as somebody who wants to discover new ways to use drugs you will do better if you uh are are able to use the tools of Behavioral cognitive science on your cells and tissues than somebody who doesn't that's that's the first claim but you could you could push it forward and say you know in our personal and and I mean I I have no credentials to be you know talking about interpersonal you know social stuff but but but overall it seems perfectly reasonable to me that we could apply the same reasoning and say um the way that uh the the same pragmatic uh reasoning to our relationships and say the way I pick frames for interacting with others is uh to see how well they work out for me so purely empirically so so if I treat you as as you know as an advanced metacognitive being we can have a certain kind of relationship and that elevates me in a certain way uh if I do something else maybe that doesn't work out as well if I take one of the lower you know kind of lower framings so again just just and it's a way to test it's it's a way to test the environment too right you we're going to go in on the assumption that this is a cognitive being and we'll see what happens and we'll see what happens I mean that's the you know you can you can go the other way you can say you can start and say okay I'm going to go on the assumption that it's not and see what the limitations are I mean I prefer the former but but but as long as as long as everybody's in agreement that this is an empirical undertaking this is not we can't we can't just have feelings about this and this is not something that you know we're gonna argue about for the next thousand years and never get anywhere this this is getting resolved because you know there's there's clear data on on a human scale of course it takes longer right so it takes maybe years for someone to say you know what my framing isn't working out for me it's I'm G to try something different right that that that may take many years but right that's kind of What's Happen happening in the whole field of plant uh intelligence that people are like let's let's let's see what happens if we assume they are cognitive or they have intelligence uh and some and some you know some people show that they can learn some people show they can't um so taking SS to the the the human Dimension which I've been looking at for this chapter I'm working on I've been talking to neuroscientists and uh you know people like Anil Seth who's has written about selves um one of the things that's curious is especially in light of what you've uh proposed in this piece is we're very invested in the idea of the unchanging self um and it's curious we don't embrace the idea that this is a completely fungible uh you know subject to our remaking and our creativity what why do you think we're invested in the idea it's it's very hard to conceive of it I mean in the definition of self is some continuity right yeah why do you think we're so invested in in the idea of a stable continuous self which you know we're being told by a lot of different uh discourses is not true I yeah I don't know I think uh you know I can I can make some hypothesis I I suspect that it's uh it's firmware left over from our evolutionary past because if if you were not um sort of wired to expect object permanence in the outside world right you know babies acquired very quickly and that they have some when they're born if if if you don't uh if if if you're not good at seeking out object permanence and if you're not uh committed to Persistence of your local self I suspect that on the Savannah and whatever the previous versions that are I suspect that doesn't play out very well you know uh it's it's not very good to say I hadn't eaten me liing you know my my patterns will continue in the universe in another form like great but but but my future self will be undisturbed my future self yeah I have I have much bigger you know bigger thoughts than this I don't I suppose that doesn't work out too well right so so I can see why why Evolution left us with some firmware that tries to kind of dumb it down to this very like basic survival um for for the same and I think I think that that firmware has a bunch of other stuff that needs to be jettisoned which is which is hard for example I think a lot of uh the issues around um you know one reason people don't like this kind of work right this this kind of uh diverse intelligence work is that they worry about false positives they worry about too many things being considered as if they were cognitive and I mean clearly you can you can make mistakes in that in that realm I have people that are in love with Bridges and married to the Eiffel Tower and stuff like this right you can that I mean that does happen but um I think what fundamentally drives this is a real deep-seated fear and a um an assumption that there's not enough love to go around literally like like like that there's a um uh it's a zero some game in terms of intelligence and that if we consider other things to be to be cognitive in some way well then mine isn't worth as much you know and and and again I I I think that's like ancient scarcity mentality firmware From Evolution that says no there isn't enough of anything to go around and there's status and and it's a zero some game and if you don't have the status in your group you know you're screwed and so that's where I think some of this is coming from um but this is all you know that's that's armchair you know armchair psychologizing I don't know yeah yeah of course the other reason I mean there are there are variables you want to keep steady right and one of the things to self does is help us with homeostasis right I mean if you if you use the model that Seth talks about or some other people um that we're process we're always processing signals from our body and we're looking not to depart from certain um you know temperature blood gases I mean there's a whole range of things that we do want to stay in a very narrow range well yes and I think that's the part of this firm that I'm talking about however um you know from the perspective of the child's body the adult body is way off on homeostatic uh you know homeostatic properties um from the perspective of the caterpillar the butterfly you know the butterfly is a terrible caterpillar like everything is off you know all of the all of the and and I think this is true in in in embryonic development you know from the from from the perspective of the of the um the you know gastro the blastula is a birth effect I in fact I think I actually think that's how development works is a it's a set of repairs activated towards a fast moving Target morphology which is encoded in bi electrics and various other things that basically what happens is you know you're you're and and we've done we've done work on this this is going to come out later later this year we have some super super cool data around this stuff that that basically development is a whole bunch of repairs I'm not even sure I believe in embryonic development anymore I think it's all regeneration I think it's all repair that basically what happens is your target is your target moves faster than the an the anatomy is trying to catch up so the target information is here the anatomy is oh my God I'm all wrong and so all these repair processes kick in and you make the metamorphosis by the time you've done that guess what the pattern has moved on again and and and you're wrong again so again you have to change right so it's a set of repair so you just keep going from stage to Stage until you catch up and that's adulthood and you sort of you know maturity and then you you sort of catch up um and then and then you get a different issue with aging but but but you know uh keeping steady yes but everything is transformation at least at least until you hit aging uh everything is supposedly a positive transformation right you're trying to you know change into whatever your next form is supposed to be do you think and I'm asking you to to speculate again there advantages to giving up on this idea of a stable self I I I think so I think I think there are major advantages well uh let's see one one one advantage is that I think that it actually has uh implications for for ethics of behavior because if if you can if you if you can sever that um sort of hardwired link between your current self and your future self then one of the things that happens quite naturally is you start to think about other beings future selves as having similar status so so normally you can be self ficient to the Future in the sense that well that's still going to be me but if you can understand that future self isn't quite the same as your current self anyway then you say well for the same reasons I care about this future self for exactly the same reason I ought to be caring about other creatures let that really exist the future selves right because because because your future self and all the hard work you're doing to prop up your future self in many ways is not as tightly connected to your current self as we think and I think breaking that a little bit would be helpful because then it would make it it would make it it would make it seem much more rational to then be be compassionate to others um you know right now right now it's very natural to invest everything in your future self but once you realize that uh you know you your your future your future self is is a different being in many ways and it's like there there are there are a number of crazy ways to um okay like like weird intuition pumps around this stuff like like here's here's here's a few examples one thing that people think about and I I think this was maybe Anthony flu's argument about reincarnation he said okay um if I don't remember if I'm not going to remember the current life then what's the difference it might you might as well just tell me that I'll be dead and some other kid will be running around somewhere on Earth and like good but there's plenty of kids running around on Earth what do I care and so so he was trying to say that that memory is is what keeps it going right and then the continuity of memory is what keeps it going but you can imagine things things like this if somebody says to you are current self um we're going to uh you need some surgery and uh you could save some bucks if you uh if you instead um instead of having real anesthesia which you just have as a paralytic so you'll be paralyzed we can do our thing and in fact I mean that's a real like that actually happens to people right that like it doesn't always work right right so so uh so then you can save you can save some bucks um the current self is going to go well absolutely not but but but but the fut your self once you come out of it and they say oh by the way oh and the other thing to know is that when you do get the anesthesia part of the components is a memory wipe that's a short-term memory wi that's also a real thing right because they don't want you remembering this stuff so so so you come out of it and somebody says to you you know what um I think that actually happened uh and uh and you know and you saved $ thousand dollars but I think you were actually there I know you don't remember a bit of it but you know and at that point you're like H all right I mean you know now now now realistically it's an issue because actually the trauma can can carry forward but let's just let's just imagine that you know let's imagine the memory wipe actually works the future self will be like yeah I mean fine good deal right so so there's an inter so so there's an interesting thing here and and I think to the extent that we kind of understand that our future self is is a construction that is going to sort of try to make sense of these memories and whatever I I think it helps you uh be be kinder to others future C because they're in the same boat so I think for that so I think for that reason it's it's useful I mean it's it's also useful more broadly because to the extent that we understand what we are and how we how the biology works and everything I think it really helps us have a more ethical relationship with other other beings because right now a lot of people uh are are walking around with this very magical view of what you know they they said the humans you know humans do this and they well which humans you know the you know 100 thousand years ago you know a million years ago like which which humans right and or or during embryogenesis where does this magic sort of kick in so so I think having a better understanding of the biology and what's actually going on is going to help us understand that gee there can be other Minds that are not the con you know the conventional modern human you know mind and and help us help us have better relationships with other systems you know both both biological and also synthetic you know artificial all that kind of stuff yeah I mean it would probably make us better people but on the other hand this idea the future self being continuous that's what gets your work done right I mean this is going to I'm I'm embarking I'm writing a book it's going to take me X years it's going to be really hard painful but my future self is going to benefit uh when it comes out uh you know when I get my Advance whatever it is um it seems to me there's a there is some adaptive value and I mean as you were pointing out on the Savannah of having a strong sense of self-continuity um even though that might make you you know an yeah yeah no no I think I mean it's it's true there there there probably is some competitive survival value to it I think that um you know look if okay I I I think back to some things I've done in the past some some let's you know some a let's say some achievements that I've done in the past yeah I I I and May and maybe it's just me I I have a um I I find it really hard to to visualize that that was me like really like I know I know historically like if if I look in the record everybody says it was me and everything so I know it's me but but but thinking back I'm like wow uh that's you know that's amazing I don't remember what it was like I don't really you know it was years ago so I don't really remember what it was like to do it and the feeling between me having done it and somebody else having done it is just a historical record at this point just a matter of historical record you know I did things before that I probably couldn't do now so what does that mean do I still keep the credit for it it's you know in in in the soet it Society to maintain records and to right and to reinforce these things but but really you know it's sort of like uh it's been a long time like do I really you know do I is it still me that did that I don't know because I don't think I could do it again you know that that kind of stuff so that's so interesting um I mean to embrace this idea of self as a as as construct and a creative act based on interpretation and misinterpretation of memories it's I mean it's a very exhilarating idea I think in many ways in that people are are very stuck on uh either the positive case for S which is a myth or the case against them which is just demystification without any sort of positive yeah benefit there's a you know there's there's there's an even um well you just just to to to put put on the radar in case in case you want to talk about it um there there's an even weirder if if you think all of that was weird weird uh the bit that I only mentioned in that paper that could that we could talk about is uh uh erasing the distinction between thoughts and thinkers because that I think that I think is even even more of a of a profound sort of yeah so talk unpack that a little because you you kind of raced over that idea and um I think it's really interesting I mean I I was actually asking this question I was talking to anneal and uh I was saying can you have a a uh a perception without a perceiver or an inference without an infer and he was saying yes you can but I had a lot of trouble getting my head around that yeah yeah let's uh so let's and so I so okay so so just to kind of fair fair warning these these ideas are pretty new so I'm still kind of chewing all this around so you know uh take everything with a with a grain of Sal but um let's uh I'll I'll start I'll start with a with a little story uh just a short this this is a real science fiction story that I read years ago and only now I'm sort of understanding the significance of but I wish I could remember whose it was I don't I don't remember uh the deal is that self down the drain yeah ex exactly right like I knew at one point I don't know um these these creatures come out of the center of the earth they're they're they're incredibly dense yeah they come out of the core they're just incredibly dense they use gamma rays for vision like super dense so they come up here and uh everything that you and I see here is solid it's just gas to them you know they see they see because because they're so dense as far as they're concerned the crust and everything up everything up here is is a very rarified kind of plasma around right and so and so one of them so they have some tool one of them's a scientist and he's got some tools and he says to the others he says you know I've been watching this gas that surrounds this planet and there are some uh there's some patterns in this gas there's some there's some persistent patterns in this gas and they look like they're doing things they they look kind of agentul and so the other one the others of course laugh at them and and they say no no look we're real you know we are actual thinkers you can't patterns can't be thinkers the patterns and gas can't be thinkers and he says no I I'm pretty like it looks like they're doing things and I've done some experiments you know and they look like they have certain cognitive properties and and then they ask well uh how long these patterns last well they only last about a 100 years well that's ridiculous I mean 100 years what what could you know what could happen you know we we live millions of years what what could happened in 100 years right so okay so so so right away and you can sort of take this this further uh you right away you start to get the idea that the distinction between a temporary but somewhat persistent pattern in a citable medium and a real solid thing that we you know this is a thing you know this is the cognizer that's a bit of in the that's in the eye of the beholder quite a bit and people who study metabolics will already say that of course we're temporary metabolic patterns that exist in there's flux that is sort of self you know people who study um uh uh you know kind of these these emergent self uh self-reinforcing energy patterns right that that's that's how they would describe us anyway so okay so so once you start thinking about that um then you can start asking about further about the distinction between thoughts as patterns within a cognitive system and what does it take to spawn off to have thoughts what does it mean to have thoughts so once you go down that road you start to think uh about imagine this Continuum so here's a here's a Continuum fleeting thoughts so these are patterns that come in and they're gone right persistent intrusive thoughts so these are thoughts that once you've had them they are hard to get rid of and not only are they hard to get rid of they do a bit of Niche construction I'm sure you you know all about this is certain kinds of thoughts actually change the brain to make it easier to keep having those thoughts right so so so they're doing a little bit of Niche construction they're um PR they're they're um modifying their environment to allow themselves to persist and multiply and and things like that you go from there and there's there's some kind of exotic intermediates but the next thing we for sure know about is um multiple personality Alters so so not just a single thought that hangs around it's like a it's like a bundle of thoughts that has some Dynamics to it and can actually you know um do do some stuff and and then and then of course there are you know things that we call full-blown personalities which are like really consistent uh at least for some period of time they're consistent sets of thoughts so so the thing about the right side of that Continuum is that these are they they've done two things one is they've closed the loop in that they um reinforce themselves so they right they they prop they they they prop themselves up over time but the other thing is that they can spawn off other thoughts so by the time you get to a to an altar you've got a you've got a cognitive um pattern that's complex enough that can spawn off other simpler patterns and so now now you try to draw a distinction between the thought and The Thinker and it's real hard now because because okay all we're talking about is cognitive patterns and some of these patterns are dissip you know in the language of dissipative system right some of them are just you know fleeting some of them are self-perpetuating like the red spot of Jupiter and things like this they sort of keep themselves going and other patterns uh spawn off smaller patterns and other patterns hang around for a hundred years and we give them names and we you know but don't they need the I mean but as you suggested they need the host of of brains right of I I don't think they well so okay so this is like very speculative stuff right so so here we go um first of all I definitely don't think need brains because these kinds of things go on in all sorts of media so they go on in in protoplasm in nor in cells in in I think I think there's lots of lots of media that they can so different substrates not not necessarily just brains exact livingres lots of different well we call the substrate living because it's a substrate that is capable of Hosting self-reinforcing goal driven patterns patterns that have agendas that's what we call living um but but I think you could have and I'm sure in the out there in the universe I'm going to guess there's all kinds of substrates that are capable of that um I think we're going to end up making a whole bunch with with different um you know both software and Material Science that people are working on I don't think there's anything special about brains per se with for that aspect but um but here's the here's the craziest thing and this is just the latest thing I thought of that I have I have zero um uh kind of uh details to back this up but I'll just throw this out as as a crazy thing uh you know at the at at the beginning of the 1900s it was thought that um self-perpetuating electromagnetic waves so a pattern that perpetuates for long periods of time required a material to to be waving you needed you needed a an an ether that you could say was something was waving in order to have a wave right we did we did away with all that we don't you you can have a pattern propagating with nothing waving there you don't need a material you got rid of The luminiferous Ether uh because because uh you found out that the two components the magnetic and the electric component you know they they can they can sort of reinforce each other and and this thing sort of sort of propagates so I wonder and again I'm not saying I believe this I'm just saying this is something I'm toying with at the moment I wonder if you could make a move like that you could um get rid of this cogniferous ether AKA brains or whatever else and you could have patterns that are actually self-reinforcing in the absence of all of that I don't know what specifically what physical model you would have I think we need to think more about this doesn't seem like you need a physical model correct the reason correct the reason you um well the the first thing that will happen is people say will say this is dualism and fair or idealism actually or or idealism yeah fair enough I I you know I I think that's true and I think some of the stuff we talked about before about um the practicalities of platonic space and those the patterns that are there like I think this is in that same in that same vein and I think yes those things are not physical and think they're real in the sense that they make a huge difference about what happens next and so on um yeah uh the reason you need some kind of a physical story at some point is that you need to flesh out uh a story about the interaction with physical bodies so what is it that happens when when Evolution produces or or an engineer produces some kind of physical machine what's the interaction between those patterns that come to resonate and so so some people will you know some some somebody will say incarnate I will say you know you know Richard Watson will say resonate uh who knows but but when it comes to when it comes to these kind of patterns that are embodied in a physical machine um we we need some kind of story about about what happens there because because otherwise you know they sort of remain off in this in this so they have to encounter something they have to yeah yeah and we're actually doing work so this is again this is not published yet um but we're actually doing some stuff on how you could BAS basically basically at this point I think that um the the idea of a of an unchanging permanent sort of platonic space where everything just is sort of fixed there I I think that I think it's much more it's much more interesting than that I think there is a chemistry of these things in that space and I think we have a computational way to start um doing experiments there and we have I have a student working on on some stuff so stay tuned there will be that it it is it is completely wild um looking at looking at agentic properties in sets of logical sentences just you know just sentences some of them are passive and some of them are not some of them do things by themselves and and it's and it's really interesting but but that's you know that's that's that's what I want to um you know going back to the whole butterfly caterpillar thing I I think that you can you can think about this thing as you've got the left side of it's it's it's a it's a Bo you know it's an hourglass SLB tie where right where you've got um you've got the the there's Intelligence on the left side that takes all of these different experiences and uh generalizes and abstracts to some sort of compressed engram you've got the creative intelligent part on the right side that has to uh reinflated into whatever the current context is and figure out what what the what the memories mean but again all of that that that first way of telling the story assumes that the information is passive and that all the intelligence is on the is in the hardware what if what if the data itself is not actually passive what if these are patterns as as we were just talking about and patterns that are using the medium of the physical process from caterpillar to butterfly right to to perpetuate and transform themselves and what if they're doing a bit of um Niche construction to uh make sure that they perpetuate and actually you know do better right as a butterfly you've got some advantages that that you can have so so I don't know you know that that goes back to our um uh the S the the self- Sorting data in the algorithms right the the Sorting algorithms where the distinction between the data and the and the algorithm is really very much in the eye of the beholder and and you can very and if you look for it you can find the data doing things you know how could you prove that that there's agency in in information well the way you prove agency in anything I think is you show uh again and this is this is an kind of an engineering take is you show how doing so helps you do something new how how right so so how how does it help you right it's it's the you know my argument is that back in the olden days when people said there's a spirit under every Rock versus now where the scientists say there isn't any anywhere both of those things are wrong because they're just you have to do experiments and you have to say here is my theory here's what it does for me you know I I i' so so what so so here's what we're doing for this um uh for the for the uh sorting algorithms thing what we found and this was just the first thing we looked for so no I I got to think there's more but but we just haven't found it yet the first thing we found is we found these algorithms doing stuff that are we that that that is not explicitly in the algorithm they are they are doing things specifically they've they're doing this clustering behavior that are not in the algorithm so there are no steps to to do that in the algorithm now now now in you know when you're looking at an algorithm it has a cost to it every computation costs something so so what we what we see is that there's an algorithm the computer's carrying out this algorithm that you pay you pay an energy toll to to do all the but it's also doing something else and it looks to me like it's doing that for free so so so here's how I would do it uh and and I have a I have a somebody in my lab who's who's who's working on this right now is to harness that to some other task that's actually useful because if you can do that that second task is getting done for free without paying an energy uh bill for it now I I mean obviously that sounds impossible I understand that and it may well turn out to be impossible but that's the kind of thing you you you you have to show an example where okay I took seriously the idea that this thing had its own go directed activity I harnessed that go directed activity to you know it's like the donkey with the carrot on the on the stick right by understanding what what the drivers of this of this agent are I can harness it to do useful work and if I can do more useful work per unit of cost for compute than somebody else who doesn't believe in that I mean that's it there's there's no more to be had than empirical success so so that's the kind of thing um that I think we're gonna have to do that's great that's fascinating so you need memories to have a self yes can you have a self without memory you can have a uh you can have a self flet you can have you can have a slice I I you know I do I do think and this is like going beyond anything that I can usefully sort of show it's just the thoughts there that um thin slices uh are uh they they do have experience and they probably do have uh you know some some sort of Consciousness associated with them but but but if you don't have carry through then then you become fleeting thoughts you become a set of fleeting thoughts basically as opposed to a coherent reinforcing pattern right that you could make a story out of yeah um so you did you just used the c-word and you used it several times in this paper um after uh last time years ago when we met you were like conspicuously avoiding it um and there's a really provocative uh sentence if maybe you could just unpack a little for me uh could Consciousness simply be what it feels like to be in charge of constant self-construction driven to reinterpret all available data in the service of choosing what to do next so tell me how you're thinking about Consciousness yeah and so and so I still I still largely avoided because what I don't have is a full-blown new Theory Of Consciousness that you know I'm prepared to defend strongly I'm working on nobody does well I mean a lot of people think they do but exactly right there's there's a lot of people who are willing to sort of get up with it get out with it I I'm I don't have anything like that yet but um but but but here's here's what did strike me uh you know um Mark SS whose work on this I really like uho has this he has this interesting um point of view that he thinks that uh he says that Consciousness is palpated uncertainty about the future so the idea is that Consciousness is the is what it feels like to have to be in charge of knowing what to do next this idea that right this so this leads to um connected to active inference and you know what what do these patterns mean what can I expect next how do I minimize my surprise these things so so so here's another hypothesis what if um yes all that is true but there's an even deeper problem which is that as a conscious agent not only do you not know what to do next you don't even know what your own memories mean you have the constant burden of having to reinterpret your own memories all the time and of course it's subconscious if it was if if this is something we had to do consciously you know we wouldn't we would never make it but it's a little bit like you know people who um have uh a brain damage and can't form new memories you know you wake up in the morning and there's a there's a there's a notepad next to your bed and it says hey guess what uh you can't form new memories here's where things stand and by the way before you go to bed tonight write write you know write write the you know write the next note and so right so I mean that's what all of us do but for for all of us that scratch Pad is internal but for ant colonies and certain kinds of patients you have to externalize you have to externalize those memories somehow so um I think that uh we we're all kind of in that boat in that every so many milliseconds you have to be able to you have to be you have to reconstruct a story of who you are what you are what your plans were right and and and we don't notice it because it's I mean it has to be automatic otherwise we notice it when we wake up in the morning yeah there there's an interesting moment of reconsolidation of self yep yep there is and and I wonder if I wonder if the this is the problem with trying to make sense of our dreams too is that and and I'm not I'm no no expert on dreams but but but I wonder if part of the issue is that at night you don't really do a super good job of memories in a way that your future self is going to decode them and that that's the the crazy Dre where you can't even figure out what what the heck it was it's basically because you're looking at these an Gams going I can't make heads of taals of this and and it's just it's just more difficult than than than waking you know waking memories I don't know that that's that's a hypothesis but I wonder you know for Consciousness I I I think that I think Mark is right but I also think that there's an even deeper task facing uh selves you know conscious selves which is that you can't even you can't even really rely on the past it's uh you know you have to you have to reconstruct it all the time you have to figure out what you are and re you know recommit to being an agent doing things so that that was that was my hypothesis yeah I think it's a fascinating idea of course and we haven't talked about the social world but that that is helping us in a funny way and and constraining us right I mean you're being told who you are all the time by the people around you you know who who kind of keep you in certain roles yeah and limit that your ability to change um yeah there's there's a softening of self that happens when you're not in a social environment it seems to me so I I thought about this too with regard to psychedelics which have interesting effects on the self and on memory um all sorts of stuff comes up people people talk about a period of ego dissolution a complete loss of sense of self yet they're still conscious um and then this kind of putting things together in a new way you know I mean especially with the trauma victims are treated um they uh they take out their memories difficult memories and when they reconsolidate them they're not the same um they've got they've lost the emotional charge or whatever it is I mean there's a kind of very interesting process of playing with your memories in psychedelic experience so that you end up being a slightly different self when you come out if it's working I mean that's kind of the theoretical model is is get you out of your habitual ways of looking at things and you're and and those kind of uh you know rumination I mean those sort of the thoughts you were describing that take up resonance in your brain that you might not necessarily want that's super interesting and I wonder if uh there is a clinical um path here by taking seriously the idea that some of those thoughts may not want to leave like literally right you know and and there may be some that is trauma I mean that's you know part of trauma yeah yeah but but but I wonder if you know taking taking serious vious ly their ability to process information maybe not in the same way as a full-blown altar but but but still not passive data but actually a pattern that can you know a A dissipative system that can actually process information you know it may I mean I don't I don't know anything about this so so maybe this is totally nuts but I've heard U various kinds of um alternative therapists and whatnot you know wanting you to talk to your different know this but but so so that may not be in that may not be crazy there right there may if if they can if these patterns can process information to some extent there may be stimuli you can give them where where basically you're not you're not just treating the brain to somehow take care of this Al that may work too but you're actually maybe maybe maybe there is something to be set for communicating well yeah this is now this is the theory in treating schizophrenia you know yes you are hearing voices why don't you talk to them or why you let me talk to them um and if you think about it trauma memories of trauma do process information because you get you know you hear a gunshot or something now and it's processed as that uh or a siren or whatever it is so they are still active in your brain in an interesting way yeah yeah that's very that's very suggestive um I'm Gonna Let You Go in a sec I just have one or two more questions um well this kind of relates to that psychedelic point but it's interesting that we we cherish ourselves and we talk about um self-esteem and building up the self you know with our kids um and we're very wetted to the idea of as I said earlier this this unchanging enduring self yet at the same time we spend a lot of effort trying to transcend it and Escape ourselves um whether it's through drugs is one way but also um travel uh experience awe that we know kind of diminish the self religion giving yourself over to the group why do you think we also want to transcend selves I think again I I don't know but I think that probably there are competing drive so so there's the there's The evolutionary firmware that says uh defend yourself at all costs right that's I think that's got to be in there but there may well be an opposing component that says yes but if you stand still you're also not going to make it so it's like um you know with active inference there's this thing you you know surprise minimization right so but but the easiest way not to be surprised is to stick your head in the corner and not look at anything and then there will be one final surprise at the end when somebody eats you but that's it until then you're not surprised so you know so so there's got to be I think in biology there's got to be two competing drives one is surprise minimization one is exploration where it says you know if you haven't seen anything new in a long time you might be stuck in a corner you better do something else right and so so maybe it's something like that maybe it's like yes defend yourself but also if you're too stuck and you haven't learned anything new and you haven't improved and you haven't whatever then um and and I don't know if that's something that's uh comes from the biology itself or meaning just you know a drive with actual um selective advantage or whether that's sort of a consequence of being an advanced being uh that you know has some sort of semis spiritual drive to um to improve over time or something right right you familiar with the D this is another binary between exploring and exploiting yeah um it kind of reminds me of that idea that there is a desire I mean we spend a lot of time in exploit mode as adults um but we also need to do the exploration and that and that's the exploit mode has the strongest sense of self yes um psychologists would say yeah yeah yeah I think that makes sense and and I wonder if I wonder if aging on the cellular level if aging is is a part of that you know well you chase you chase the the the moving Target morphology as an embryo and eventually you reach your adult form and sort sort of that's it you've caught you've caught up and so now the name of the game is to defend against tiny local um defections of cancer and degeneration and and just you know cells getting old and so on but but but if once you've given up uh once you've given up that you're on a trajectory for some kind of construction project maybe that you know that that's it if you if you're if you're stuck in a rut of exploit and you're really not exploring anything that may be when things start to go to pot I I don't know I mean you know the the the kind of uh familiar folk example is you know the people who who retire from from their job and then immediately drop dead right so that's the right that that's just kind of a but but maybe something like that is going on um on a on a on a cellular level where basically you know uh when there's nothing pulling you there there is no more of that uh the forces which is what we study during development that pull you from stage to stage at that point then then then things things start to go downhill I don't know yeah um I noticed in your um acknowledgements which was a fascinating list I read it um Bernardo castr was in there yeah tell me about I'm talking to him tomorrow I've never met him oh yeah yeah we've yeah we've had a i' I've had a couple of conversations with with Bernardo um online and you know a bunch off uh yeah super interesting dude um computer science uh kind of background philosophy um you know he's got this kind of idealist model um yeah I think he's very interesting I you know he did a he did a cool uh session with rert spir and they sort of talked about this notion of uh all of us being dissociative altars of the great Cosmic mind and things like that um yeah I I I think you'll have fun I think he's he's he's a very smart guy I think you'll have fun I I find I'm reading his book right now he has a new book that he sent me a copy of um the epistemology piece seems very persuasive I mean the more we learn about active inference and the fact that we're constructing our perceptions are limited by our cognitive equipment and the world we're seeing is mental to a greater extent than we realize um but then he goes to the next step that what's really there is mind and that's I don't understand that he sort of asserts it without arguing it um well he's got he's got I think five other books that yeah I know he's probably leaning on and I'm not I'm not going to try to um kind of make his argument for him because I think I'll butcher it he he'll he'll do a much better job of it himself you can ask him to just like say it but uh you know I I think that fundamentally I I I I like that idealist position in the sense that I I think what actually propagates through Evolution Through the Universe I I think the the fundamental um pieces are are perspectives so I I don't think it's genes I don't think it's bodies I think it's um I think it's uh I think it's perspectives and so well what I mean is that that any any agent uh you C you can't afford to um you can't you can't afford to try to see the world as it is in the sense of like a lassan demon you can't afford to track micro States because if you if you try to track micro states by the time you figure out what's going on you'll be long dead any any biological system that in in the real world has constraints of energy of time uh so so what that means is you're going to core screen you're going to decide what am I ignoring what am I paying attention to I'm going to take all these kinds of observations and I'm just going to lump them all together and I'm going to call you know and I don't mean this is not just humans doing this is every cell every bacterium everything's doing this so what you really have there then is a perspective it's a commitment to it's a set of choices that say here's what I'm paying attention to and here's um uh how I make sense of it here's what I think it means you know it's a it's a it's an interpretation so so that's what I think is perpetuating through the universe and being selected on and you know altering with time and all that is is a set of perspectives a set of a particular vantage point of saying here's how I make sense of my world and and once we understand that that every cognitive being is limit every finite every real being is finite we're all limited we all have to choose a perspective that ignores all kinds of other stuff and it's not just about only being able to sense a small um narrow band of the spectrum it actually I mean that's true too but but it's much more than that it's cognitively not just perceptually that we're limited uh then I think uh you know then then yeah then it's then it's all about uh what we're are you talking about perspectives in the in the in the minds of living things or are you talking about perspectives at large I'm talking about perspectives I mean I I we can talk about what living things are but I think that any agent that has to whether we would call it living or not I mean any any kind of robotic thing any kind of a uh system that solves problems in in a world of some sort is going to have to have a perspective on that world and the more right for a rock you don't worry too much about its perspective although you with potential energy and things like that you almost kind of get there but but but for most things things that have a significant inner perspective we call those things living typically but doesn't have to be uh and and yeah and whatever World they you know you can you can ask like Gene regulatory networks what's the world that they navigate it's not the physical threedimensional world it's the it's a high-dimensional transcriptional space that they live in and I think from that P from from no pun intended from that perspective what you put what what ends up being emphasized a lot more is what choices are you making about your uh cognitive um uh framing of your world and a lot less about some uh objective World existing out there somewhere and uh yeah I think I think that's how you get to idealism is just is just by recognizing that all um uh real real agents are going to be finite and limited therefore what you're really talking about is uh an inner world that you're building for yourself it's it's your own self model that's that's that's at play here not not some objective thing and that your self model May or your model of the world may or may not be compatible with mine um you know yeah so in your view though I'm I'm assuming something but I could be wrong um the prerequisites of self and the and and the process you're describing do you think that that can be embodied in a computer yes but computer has to I mean again whether something in a is a computer is in the eye of the beholder what what what we what what we mean when we say something is a computer is that I have a set of uh conceptual tools that I bring to the table here that come from computer science and I'm claiming that those tools are useful that that's all that's all it means and um do I think that uh some of those tools are useful to understand cognition yes you know for example reprogrammability and software I think it's extremely useful uh do I think the vanoyan architecture and the sharp distinction between data and the Machine are particularly useful no I think we'll have to dump those uh do you know do I think that at some point we could make synthetic uh beings that have the right features absolutely I don't see any reason why uh it it see it seems completely backwards to me to say that the the blind uh meanderings of The evolutionary process can create real minds but but but intelligent Engineers who understand how minds work can't do it that that seems crazy to me so so I I think that that for sure once we understand and I think we're getting there I think we understand you know a few of the key features um I I don't see any reason why we couldn't uh reimplement them in a synthetic substrate sure yeah but you need a substrate that as youve said is not at all like a van v noyman correct correct I yeah I think you need a number of things and the and the thing is that um I I started I started WR those things down I started writing a paper this was months ago I started writing a paper to say look these are the things we've learned from biology if you just do these things you will have you know a real a real agent and then then I stopped and and although I don't think this is going to solve it because somebody else will of course do it but I didn't want to be responsible for it because I think that to whatever extent that's on the right track it may not be but but if it is then what that means is that it makes it very easy to make trillions of other beings that are have um that have more that deserve moral consideration and I'm not really interested in being the cause of that but I'm sure that's where we're going I'm sure I'm sure this will get figured out and this is what will happen yeah other people are working on it oh yeah including including Mark Zs by the way yes yes that's that's right that's right and I would very simplistic and it's and it's using you know computers as as they now exist yeah I think that um I think of of of anybody's uh project that uh that I think has a chance of doing something like that I think his is at the top of the list I actually think he could do it yeah amazing amazing well thank you are you writing a book a g a trade book uh I am uh I'm currently um on Pagan and I are writing a book on bioelectricity and uh it doesn't it doesn't have any of this stuff in it it's just straight up bioelectricity uh but the next book if I you know if I if I uh get there uh the next book will be all of this stuff basically great Evolution cognition all that stuff and are you going to do that for a general audience I don't you know I don't know I I'm I'm still we're still um so so Richard Watson and I are probably going to do it together or or maybe we'll do two books we we haven't been able like maybe it'll be two books back toback or something we'll have you know maybe inverted like the way that uh um uh it was done for the planaria journal and and and all that uh I don't I don't know um I I'm I'm having a really hard time figuring out what what is the best thing to do here because on the one hand I could do a book for for a general audience and I think what would happen then is that a lot of these there there will be a set of people from let's say the alternative sort of um you know new age kind of community who would read these things and they say oh yeah no kidding we we we've known this for a long time and we you know they don't they don't need you know they don't need um you know scientific you know sort of experiments to prove it whatever they they're already in in for it um and then there will be the science Community who will say well there's not enough uh meat here to convince me of any of these crazy things right so so I so I worry about that issue for the tradebook you know um I it sounds like that could miss uh the impact on the other hand on the other hand if we do a you know an academic press book where you where you go into detail on on all of this stuff uh yeah I'm not you know I'm not sure I'm not sure which of my colleagues have time to read something like that uh you know who who's read it I don't know so so I still haven't figured out what the right I mean I think you've got a really interesting book around some of these ideas and um uh I think it'd be of interest to a lot more than new age people um I really do um do you have a literary agent I do yeah I do yeah um I mean it's a conversation to have with that person is it who is it brockman's shop or it is yeah yeah yeah Dan introduced me to Brockman yeah yeah yeah we'll have a I mean I mean he obviously wants a trade book all right so um now the other thing is you know they don't which which is kind of wild they don't like um images and they certainly don't like color I have I have some amazing uh color illustrations and I don't know that feels weird to me in this modern day and age with computer like do we even need a book I couldn't I just put it all on a website like why do we need a book I don't you know you know and somebody's gonna tell you not to use color like I don't know yeah um yeah well good luck with that I I I just think that these ideas would be of interest to a lot of people in a lot of different fields they're very interdisciplinary ideas and um so uh I think the right trade book would reach your colleagues or colleagues and other you know in philosophy and computer science um anyway it's all it's just fascinating stuff well thank you I appreciate that I'll I'll be in touch because because I will definitely want to pick your brain when it comes time happy too I'm happy to read a proposal if you do that or whatever kind of suggestions um that be super useful thank you good and I'm gonna be in Cambridge this fall so uh I'll look you up we can get together in person definitely yeah we'll hang out